The Glove Problem

Spring 2022 ARML Power Contest

The Background

In 1977, the puzzle master Martin Gardner published a puzzle in *Isaac Asimov's Science Fiction Magazine* about doctors who need to operate on patients while trying to avoid the Barsoomian flu. The only problem is that they do not have enough gloves for each doctor-patient encounter to use a pair of sterile gloves. This Power Contest will explore the puzzle and some generalizations. (The word "Barsoomian" was a nod to another science fiction author, Edgar Rice Burroughs.)

In the original version of the problem, there were three doctors and one patient. Each doctor had to perform an operation on the patient, but there were only two pairs of surgical gloves available. Since the flu transferred by touch, no doctor or patient could be exposed to a surface of a glove that had previously been exposed to a different person. Their solution: Doctor X wears *both* pairs of gloves and operates. Doctor Y then wears just the pair of gloves that touched the patient—the inside of those gloves are still safe because Doctor X wasn't touching them! Doctor Z then takes the gloves that were worn by Doctor X, *turns them inside-out* so as to wear them while touching the remaining safe side, and then places the (now out)side, possibly infected by Doctor X, inside the gloves that have touched the patient to perform the final operation.

We will use the following notation. There are m doctors, denoted $D_1, D_2, \ldots, D_m$, and n patients who are denoted $P_1, P_2, \ldots, P_n$. When a problem involves using k pairs of gloves, the pairs will be denoted $G_1, G_2, \ldots, G_k$. A pair of gloves that has been turned inside-out will be denoted $\overline{G}$. We list the doctors, gloves, and patients involved in a particular operation in that order. So the three operations in the solution to the original problem would be listed as:

1. $D_1\, G_1\, G_2\, P_1$
2. $D_2\, G_2\, P_1$
3. $D_3\, \overline{G_1}\, G_2\, P_1$

A variation of the problem has two doctors and two patients. Even though there are now *four* operations taking place, only two pairs of gloves are needed, and they do not even have to be turned inside-out! Doctor X wears both pairs 1 and 2 of gloves and operates on A. The outer pair 2 which have been contaminated on the outside by A are worn by themselves by Doctor Y to operate on A. Meanwhile, Doctor X operates on B wearing just pair 1 (whose outsides are still clean), touching B only with that clear side. Finally, Doctor Y puts the comtaminated-by-A outsides of pair 2 into the comtaminated-by-X insides of pair 1 to operate on B—who is still only being touched by the outsides of pair 1.

You should make sure you understand the notation of this problem by notating the solution just given to the two-doctor-two-patient problem; an answer will be given at the end of this reading section.

In the problems that follow, a *protocol* is a list of operations, in the order they are to be performed, showing which doctor, which patient, and which pair(s) of gloves are used, noting if any of the pairs have been turned inside-out. You should use the doctor-patient-glove notation as outlined above when needed. You may also use other standard mathematical notation, including the ceiling and floor functions, $\lceil x \rceil$ and $\lfloor x \rfloor$, as well as functions that choose the minimum or maximum of their inputs, $\min(x_1, x_2, \ldots, x_n)$ and $\max(x_1, x_2, \ldots, x_n)$.

You may invent other notations if convenient. If you do so you must either: 1) define the notation in each problem where it is used, or 2) include a separate sheet at the front of your solution packet where you gather all notations used and define them for the entire packet. This page should not include the solutions to any problems—notation definitions only!

For reasons that will become clear, a pair of gloves cannot be separated into two individual gloves. That is, both gloves from a pair must be worn for an operation, and if a doctor is wearing multiple pairs, each layer must consist of both gloves from the same pair. Furthermore, either both gloves are turned inside out or neither are. This justifies the notation using a single letter to represent a pair of gloves, rather than having to keep track of the left and right gloves separately.

The following is a fairly efficient protocol for $n = 2x$ patients and $m = 3y$ doctors, that uses exactly $x + 2y + 1$ pairs of gloves. Make sure you understand how this protocol works, as it is a model for some of the protocols you will need to construct in the problems.

For convenience, the gloves will be denoted $G_1, \ldots, G_x$, $H_1, \ldots, H_y$, $I_1, \ldots, I_y$, and J. Start by dividing the patients into pairs P_1 and Q_1, P_2 and Q_2, …, P_x and Q_x. Divide the doctors into trios, A_1-B_1-C_1, A_2-B_2-C_2, …, A_y-B_y-C_y.

Step 1: Give the pair G_i of gloves to patient P_i for i from 1 to x. Give pair H_j to doctor A_j, and pair I_k to doctor B_k for $1 \leq j, k \leq y$. Now each A- and B-doctor can operate on each P-patient via the operations $A_j\, H_j\, G_i\, P_i$ and $B_k\, I_k\, G_i\, P_i$. The insides of all H- and I-gloves are now contaminated and can only be touched by the doctor who wore them for this step. The outsides of each G-pair are contaminated by one patient who is the only one who can use that side from now on. The insides of the G-gloves and the outsides of the H- and I-gloves are still clean, though, and can be used by anyone.

Step 2: The I-gloves are turned inside-out and given to the C-doctors. The C-doctors now operate on the P-patients, using pair J to protect the inside of the G-gloves. That is, perform the operations $C_k\, \overline{I_k}\, J\, G_i\, P_i$ for all i and k. While performing these operations, the inside of glove J becomes contaminated by touching many different contaminated gloves, so that side as well as the (now) outsides of the I_k which have touched the inside of J can't ever touch any person. Meanwhile, the outside of the H-gloves and the inside of the G-gloves are still clean, and the P-patients have been operated upon by all the doctors.

Step 3: Since the P-patients are done and no longer need their gloves, turn them inside out and give them to the Q-patients. The C-doctors can now operate on all these patients, via the operations $C_k\, \overline{I_k}\, \overline{G_i}\, Q_i$. The C-doctors are now finished operating.

Step 4: Notice that the outsides of the H-gloves are still clean. Protect them with the clean outside of J, by performing all the operations $A_j\, H_j\, \overline{J}\, \overline{G_i}\, Q_i$.

Step 5: Since the outsides of the H-gloves are clean, turn them inside-out and give them to the B-doctors. Perform all the operations $B_k\, \overline{H_k}\, \overline{G_i}\, Q_i$.

Notice that one side of J remains clean even though the other side has been contaminated by many different patients and doctors. Notice also we could even have started with one side of J already contaminated and it would have made no difference.

A protocol determines a diagram as follows. A diagram consists of points and arrows that connect the points (the arrows need not be drawn as straight lines). The points correspond to all the people, doctors and patients, involved in the protocol. The arrows correspond to gloves, and connect the first person who contaminates one side of the gloves with the person who first contaminates the other side.

More specifically, the first time a pair of gloves is contaminated (which may be by direct contact with a person, or by being placed in contact with an already-contaminated pair of gloves), an arrow starts at the point representing that person. If the contamination is with several people simultaneously (for example, the gloves are placed inside another pair which have already had several other contaminated pairs placed inside of them), then one of those people is chosen at random to be the beginning of the arrow. When the second side of the gloves become contaminated, the arrow ends at the point representing the person by whom it was contaminated. Again, if several contaminations occur simultaneously, any one of the people may be chosen. If the second side never becomes contaminated (like pair J above) then the arrow returns to the person with whom it started. Finally, if both sides of the gloves are contaminated for the first time simultaneously, the direction of the arrow can be chosen at random.

The diagram determined by the three-doctor-one-patient protocol is shown below:

$$\begin{tikzcd} X \bullet \arrow[r] & \bullet Z \\ Y \bullet & \bullet P \arrow[l] \end{tikzcd}$$

Be sure you can draw diagrams corresponding to protocols. The diagram for the two-doctor-two-patient protocol is given at the end of this material.

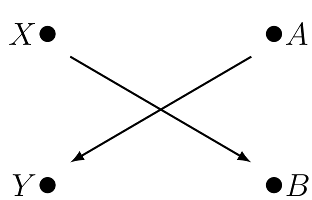
You may use the language of directed graphs when discussing diagrams, if you feel comfortable doing so.

One final concept needs introduction. For a given protocol, its *makespan* is the amount of time it takes to complete it, allowing operations to take place simultaneously if they do not involve any of the same doctors, patients, or pairs of gloves. For the sake of convenience, assume all operations take exactly one hour and no time is needed between operations for patients, doctors, or gloves. The makespan of the protocol that solves the original problem is 3 hours, since each of three doctors operates in succession on the patient. The makespan of the 2-doctor-2-patient problem is also 3 hours. This is because the middle two operations can be performed simultaneously. The makespan of the protocol for $2x$ patients and $3y$ doctors depends on x and y and is quite long, as several steps require pair J to be used in each operation!

The notation for the two-doctor-two-patient solution given in the text is:

1. $D_1\,G_1\,G_2\,P_1$
2. $D_2\,G_2\,P_1$
3. $D_1\,G_1\,P_2$
4. $D_2\,G_2\,G_1\,P_2$

This is one way to draw the corresponding diagram.



The Problems

There are 40 points available on this contest. The numbers at the end of each problem are the number of points that problem is worth. Be sure you have read and are familiar with the background material. Note especially the instructions about notation and vocabulary given in that material

1. When m doctors must each operate on each of n patients, explain why at least $\left\lceil \frac{m+n}{2} \right\rceil$ pairs of gloves are needed, given the restriction noted in the background material that pairs of gloves cannot be separated. Recall: $\lceil x \rceil$ is the *ceiling* of x, the least integer greater than or equal to x. (Hint: think about why one pair of gloves cannot suffice for three people, or two pairs for five people.) [2 pts.]

Let's see why we need the restriction that pairs of gloves must be worn together. Let there be t total people, doctors and patients combined. Number them $A_1, A_2, \ldots, A_t$, without regard to whether they are doctors or patients. Give each person a right glove (which could be a left glove turned inside-out), and number the gloves according to the person to whom it is given so that person A_i starts by possessing right glove G_i. Then, if person A_i needs to operate on patient A_j, A_i can wear gloves $G_i\, G_{i+1} \ldots G_{j-1}$ on their right hand. If $j < i$ then after glove G_t is reached the next layer will be glove G_1. That is, the sequence $i, i+1, \ldots, j-1$ is treated modulo t.

2. What glove(s), and in what order, can person A_i wear on their left hand to operate on A_j? [2 pts.]

Note that with this scheme, only t gloves—$\lceil t/2 \rceil$ pairs of gloves—are needed, achieving the lower bound proven in problem 1. In fact, if this scheme is employed, all the doctors could also be patients, and *vice versa*, and each person could operate on any collection of other people, any number of times, with the operations between various people occurring in any order. This obviously makes the problem somewhat trivial, and engenders our requirement that both gloves be worn as a pair.

3. According to problem 1, at least three pairs of gloves are needed when three doctors need to operate on two patients, each doctor operating once on each patient. Describe a protocol that allows each doctor to operate on each patient with no one ever touching a side of the gloves that already touched someone else, using exactly three pairs of gloves. Make sure to notate your solution in the notation given in the background material. [3 pts.]

The goal of this problem set is to find a protocol for m doctors and n patients that requires the fewest possible number of pairs of gloves.

4. Explain why the problem is *symmetric* in doctors and patients. In other words, if there is a protocol for m doctors and n patients using only k pairs of gloves, show that there is a protocol for n doctors and m patients using k pairs of gloves. [2 pts.]

5. Assume that every operation takes exactly one hour, that no doctor may operate on more than one patient at a time, and that no patient may be operated upon by more than one doctor at a time. Operations on different patients by different doctors using different pairs of gloves can be performed at the same time.

   (a) In terms of x and y, determine the makespan of each of the five steps making up the protocol for $2x$ patients and $3y$ doctors given in the background material. Assume that operations from different steps may *not* be performed simultaneously. (There should be five separate calculations, one for each step, in your answer to this problem.) [5 pts.]

   (b) If, instead, there are $2x+3y$ pairs of gloves available, each doctor and each patient can receive their own pair. Then any doctor can operate on any patient at any time, as the outsides of the doctors' gloves and the insides of the patients' gloves never become contaminated and no exchanges of gloves need to take place. Compute the makespan for completing all operations under these conditions, assuming as many operations take place simultaneously as possible. [2 pts.]

6. Recall that the protocol given in the introductory notes allows for $2x$ patients and $3y$ doctors, and uses $x + 2y + 1$ pairs of gloves. Create a protocol that allows for $2x$ patients and $3y + 1$ doctors, still using $x + 2y + 1$ pairs of gloves. [3 pts.]

7. Explain how to modify your protocol from problem 6 to allow for $2x$ patients and $3y+2$ doctors, using $x + 2y + 2$ pairs of gloves. That is, allow for one extra doctor by using one extra pair of gloves. [1 pt.]

8. Explain how to modify your protocol from problem 6 to allow for $2x + 1$ patients and $3y + 1$ doctors, using $x + 2y + 2$ pairs of gloves. That is, allow for one extra patient by using one extra pair of gloves. [1 pt.]

There is a very complex protocol that allows $2x$ patients and $3y$ doctors to use only $x+2y$ pairs of gloves, thus saving one pair from the protocol given in the background material. It will be given in the solution packet but you are not being asked to find it. It can be modified to show that $2x + 1$ patients and $3y$ doctors have a protocol using $x + 2y + 1$ pairs of gloves, and that $2x + 1$ patients and $3y + 2$ doctors have a protocol using $x + 2y + 2$ pairs of gloves.

9. Show that, given the information in the above paragraph and preceding problems, there is a protocol for m doctors and n patients using $\lceil \frac{2}{3}m + \frac{1}{2}n \rceil$ pairs of gloves. [2 pts.]

10. Show that there is also a protocol requiring $\lceil \frac{1}{2}m + \frac{2}{3}n \rceil$ pairs of gloves. This could be fewer pairs if $m > n$. [1 pt.]

Having determined an upper bound for the number of pairs of gloves, we proceed to prove this is optimal by showing that it is also the lower bound. The proof will involve the diagrams corresponding to the protocol.

11. Recall the protocol given in the introductory material for $2x$ patients and $3y$ doctors using $x + 2y + 1$ pairs of gloves. Draw a diagram for this protocol when there are 4 patients, 6 doctors, and 7 pairs of gloves. [3 pts.]

12. Explain why, for any protocol allowing all doctors to operate on all patients, every point in the corresponding diagram has at least one arrow coming out of it or at least one arrow pointing into it. [2 pts.]

A *component* of a diagram is a set of points within the diagram for which you can travel from any point in the set to any other point in the set by traveling only along the arrows of the diagram (you may go in either direction along an arrow—you do not have to follow the direction the arrow is pointing). In the diagram for three doctors and one patient in the background material, there are two components: $\{X, Z\}$ and $\{Y, P\}$.

13. Let $s \to t$ be a component of the diagram. That is, there is a single arrow from s to t, and no other arrows going into or coming out of either s or t. Prove that any operation involving s but not t must occur before any operation involving t but not s. [3 pts.]

A component such as the above, with exactly one arrow connecting s and t and no other arrows beginning or ending at either s or t will be called a 2-component.

14. Consider a protocol in which every doctor operates on every patient.
    (a) Prove that the diagram for this protocol cannot both a 2-component connecting two doctors and a 2-component connecting two patients. [2 pts.]
    (b) Prove that the diagram for this protocol has at most two 2-components connecting a doctor and a patient. [2 pts.]

Notice that in a 2-components there are exactly half as many arrows (1) as points (2). A component with just one point has to have at least one arrow by problem 12, so the ratio of arrows to points is at least 1. In any component of two points that is not a 2-component there must be at least one more arrow connecting the two points, and again the ratio of arrows to points is at least 1. In a component with n points, there must be at least $n - 1$ arrows holding them together. So the least that the ratio of arrows to points could be for a component with 3 or more points is 2/3.

15. Assume that there are n patients and m doctors with $n \geq m$, $n \geq 2$ and $m \geq 1$. Prove that at least $\lceil \frac{1}{2}n + \frac{2}{3}m - \frac{1}{3} \rceil$ pairs of gloves are needed for any protocol that allows every doctor to operate on every patient with no contamination. [3 pts.]

16. Under what conditions on m and n is $\lceil \frac{1}{2}n + \frac{2}{3}m - \frac{1}{3} \rceil = \lceil \frac{1}{2}n + \frac{2}{3}m \rceil$?

It is known that $\lceil \frac{1}{2}n + \frac{2}{3}m \rceil$ (assuming $n \geq m$) pairs of gloves is the least number of pairs of gloves that can be used when m doctors must each operate on each of n patients. The last problem showed that this was the lower bound in all but a few cases. Just a little extra work is needed (you have the tools to complete this work, but it involves a little more casework than most teams have the time to complete). Problem 6 and those following it had you find protocols that fulfilled this lower bound in some cases, and the complex protocol and its variants alluded to there fulfill the lower bound in the rest of the cases. Again, the details of the more complex protocol are more than most teams have time to work with, but following the ideas contained in these problems you may be able to construct that protocol. It will be given in the solution set. The protocol can even be extended to allow all the doctors to operate on each other as well, without needing any extra pairs of gloves. More history will be given in the solution set!